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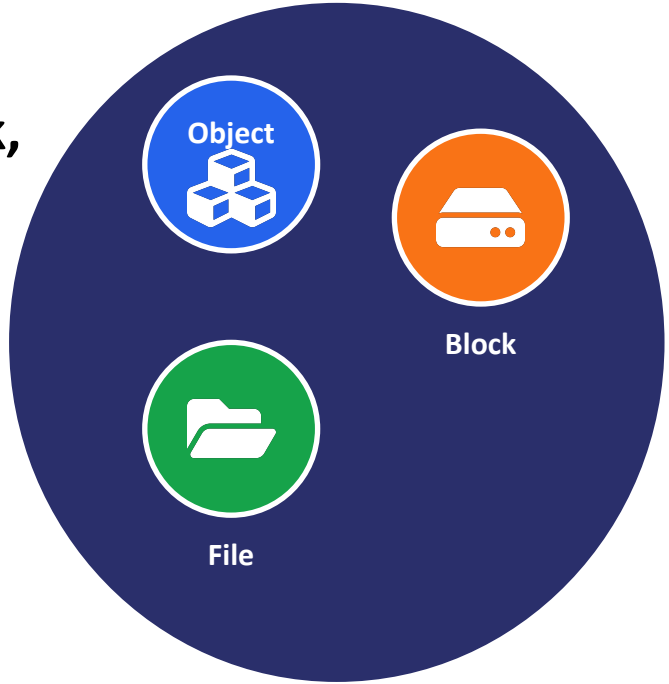
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Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Cloud Storage Technologies: Object, Block, and File Storage Compared

Compare storage architectures, use cases, and industry adoption patterns.



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Why Cloud Storage Architecture Matters

- Data volumes are growing rapidly, driven by analytics, AI/ML, media, and IoT workloads.
- Cloud providers offer three core storage models: object, block, and file.
- Each model organizes, addresses, and serves data differently.
- Choosing the wrong model leads to higher cost, poor performance, or limited scalability.
- Most real-world systems combine two or more storage types.



Three core storage models

Block Storage

Block storage is a method of storing data by dividing it into small, fixed-size blocks. Each block has its own address. It provides fast data access and is commonly used for applications that require high performance.

Advantages:

- High speed
- Low latency
- Suitable for frequent read and write operations

Uses: Databases, virtual machines, and banking applications.

Examples: Amazon EBS, Azure Managed Disks, and Google Cloud Persistent Disk.



Fast raw volumes for servers

Object Storage

Object storage is a cloud storage method that stores data as individual objects. Each object contains data, metadata, and a unique ID. It is suitable for storing large amounts of data at a low cost.

Advantages:

- Low cost
- High scalability
- High durability

Uses: Photos, videos, backups, and AI datasets.

Examples: Azure Blob Storage, and Google Cloud Storage.



Objects + metadata in flat buckets

File Storage

File storage is a method of storing data in files and folders using a hierarchical structure. It allows multiple users to access and share files over a network.

Advantages:

- Easy to organize files
- Supports file sharing
- Familiar folder structure

Uses: College documents, office files, shared folders, and content management.

Examples: Amazon EFS, Azure Files, and Google Cloud Filestore.



Shared folders and files

Typical Use Cases

Object

- Backup and archive
- Data lakes and analytics
- Media and web content
- AI/ML training data

Block

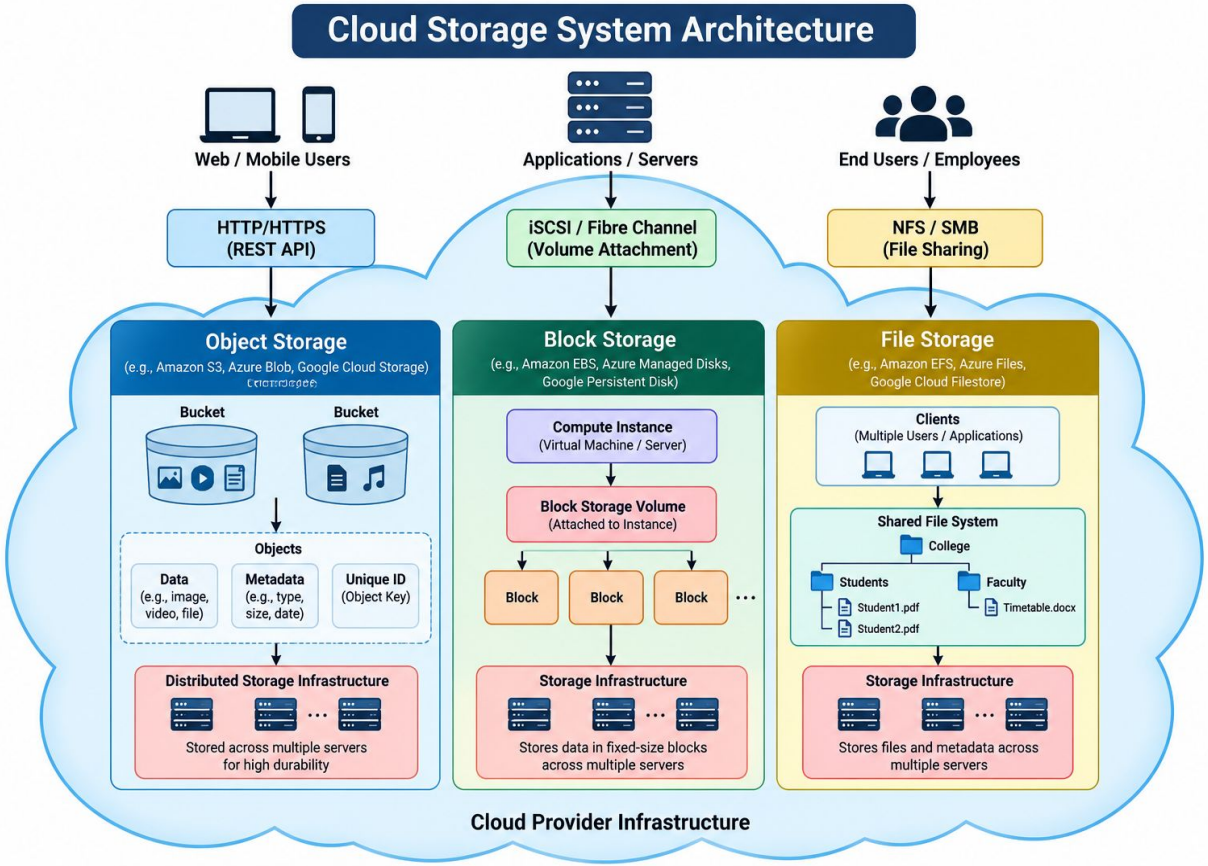
- Databases (OLTP)
- Virtual machine disks
- ERP and transactional apps
- Boot volumes

File

- Shared home directories
- Content management
- Dev/test environments
- Lift-and-shift apps



Match workload to storage


Cloud Storage System Architecture
Web / Mobile Users
Applications / Servers
End Users / Employees
HTTP/HTTPS (REST API)
iSCSI / Fibre Channel (Volume Attachment)
NFS / SMB (File Sharing)
Object Storage
(e.g., Amazon S3, Azure Blob, Google Cloud Storage)
Bucket
Bucket
Objects
Data (e.g., image, video, file)
Metadata (e.g., type, size, date)
Unique ID (Object Key)
Distributed Storage Infrastructure
Stored across multiple servers for high durability
Block Storage
(e.g., Amazon EBS, Azure Managed Disks, Google Persistent Disk)
Compute Instance (Virtual Machine / Server)
Block Storage Volume (Attached to Instance)
Block
Block
Block
Storage Infrastructure
Stores data in fixed-size blocks across multiple servers
File Storage
(e.g., Amazon EFS, Azure Files, Google Cloud Filestore)
Clients (Multiple Users / Applications)
Shared File System
College
Students
Faculty
Student1.pdf
Student2.pdf
Timetable.docx
Storage Infrastructure
Stores files and metadata across multiple servers
Cloud Provider Infrastructure

Conclusion

- Cloud storage helps store and manage data.
- **Object Storage** is used for photos, videos, and backups.
- **Block Storage** is used for databases and fast data access.
- **File Storage** is used to organize and share files and folders.
- Each storage type has different advantages.
- Choosing the right storage improves performance and reduces costs.
- Cloud storage makes data storage easier and more flexible.

